ENGINEERING DESIGN AND ANALYSIS OF THE ARIES-CS POWER PLANT

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The ARIES-CS team has concluded an integrated study of a compact stellarator power plant, involving physics and engineering design optimization. Key engineering considerations include the size of the power core, access for maintenance, and the minimum distance required between the plasma and the coil to provide acceptable shielding and breeding. Our preferred power core option in a three-field-period configuration is a dual-coolant (He + Pb-17Li) ferritic steel modular blanket concept coupled with a Brayton power cycle and a port-based maintenance scheme. In parallel with a physics effort to help determine the location and peak heat load to the divertor, we developed a helium-cooled W alloy/ferritic steel divertor design able to accommodate 10 MW/m^2. We also developed an intercoil structure design to accommodate the electromagnetic forces within each field period while allowing for penetrations required for maintenance, plasma control, coolant lines, and supporting legs for the in-vessel components.

This paper summarizes the key engineering outcomes from the study. The engineering design of the fusion power core components (including the blanket and divertor) are described and key results from the supporting analyses presented, including stress analyses of the components and thermal-hydraulic analyses of the power core coupled to a Brayton cycle. The preferred port-based maintenance scheme is briefly described and the integration of the power core is discussed. The key stellarator-specific challenges affecting the design are highlighted, including the impact of the minimum plasma-coil distance, the maintenance, integration, and coil design requirements, and the need for alpha power accommodation.

KEYWORDS: *compact stellarator, fusion power plant, design integration*

Note: Some figures in this paper are in color only in the electronic version.

I. INTRODUCTION

The ARIES-CS team has conducted a three-phase integrated study of a compact stellarator (CS) power plant, involving design optimization through trade-offs among a large number of physics parameters and engineering constraints.[1] The first phase of the ARIES-CS study was devoted to the initial exploration of physics and engineering options, requirements, and constraints. Several compact stellarator configurations were considered, scoping analyses of plasma parameters were performed, and different possible coil topologies were studied including two- and three-field-period options (based on MHH2 and NCSX, respectively[2]). Key considerations impacting the design of the compact stellarator include the size of the reactor, access for maintenance, and the minimum

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distance between plasma and coil that affects shielding and also breeding if sufficient blanket coverage is not provided. These considerations were included in the engineering effort during that phase to scope out different maintenance schemes and power core designs (including a number of different blanket concepts) best suited to a compact stellarator configuration.[3]

The second phase of the study was focused on the exploration of the configuration design space and on trade-off studies using the system code to converge on the more attractive sets of parameters for a power plant. The engineering effort included a detailed assessment and optimization of the most attractive blanket configurations, maintenance schemes, and coil structure designs to help select an appropriate combination for the detailed design study performed during the final phase of the study. The reference blanket option is a dual-coolant concept with self-cooled Pb-17Li zones and He-cooled ferritic steel (FS) structure. A higher-performance, higher-risk option with self-cooled Pb-17Li and SiC_f/SiC composite as structural material is maintained as a backup option. A key part of the overall engineering effort included an optimization of the blanket and shield in local regions where space between the plasma and coil is constrained to downsize the radial build and to allow for more compact machine sizes.[4]

Two major maintenance schemes were considered during the second phase of the study to provide a broad range of possibilities to accommodate the physics optimization of the number of coils and the machine size before final convergence for the integrated design study intended for the final phase of the study:

1. port-based maintenance, chosen as the reference scheme for the final phase of the study, whereby replacement of the blanket modules is done through a limited number of designated maintenance ports
2. field-period maintenance, with replacement of integral in-reactor unit(s) based on a field period, including disassembly of the modular coil system.

The choice of maintenance scheme affects the reactor layout and the design of the in-reactor components. For example, for the selected port-based maintenance scheme, the vacuum vessel is internal to the coil and the replaceable units are modular. For field-period–based maintenance, the vacuum vessel would be external to the coil and there would be less of a size constraint on the replaceable in-reactor components.[5]

The engineering effort also focused on developing a divertor design that can accommodate the expected heat flux and a coil configuration and structural design that can accommodate the forces on the coil. One key plasma-facing issue is the accommodation of the rather high alpha-particle loss fraction.

The following sections describe and discuss in more detail the engineering design of ARIES-CS along with a summary of the key results from the detailed nuclear, stress, thermal-hydraulic, and power cycle analyses performed in support of the design. Key stellarator-specific challenges affecting the design are highlighted, including the impact of the minimum plasma-coil distance, the coil design requirements, and the need for alpha power accommodation. The machine, power, and nuclear parameters are first summarized. The maintenance scheme is then described. Next, details of the blanket concept are provided, including its coupling with the power cycle. The divertor design is then described and the impact of the alpha-particle loss discussed. The coil configuration and structural design are then summarized and concluding remarks made. It should be noted that this paper is intended to highlight the overall engineering effort and the integrated and interrelated effects of designing the different machine components. More details on the specific design and analysis of the different components can be found in companion papers, as referenced throughout the text. Also, the analyses presented here and in the other papers were carried out at different times during the study with somewhat different system parameters. To save time and resources, it was decided that analyses would be redone only when the slightly different base parameters (as compared to the final system study runs) could significantly affect the final results and findings. Thus, parameters presented here and in other papers might slightly differ but would not affect the overall conclusions.

II. POWER PLANT PARAMETERS

II.A. Machine Layout and Power Flow

Optimization of the power plant parameters proceeded through a combination of physics,[6] engineering, and system studies.[7] The reference three-field-period configuration considered during the final phase of the study is illustrated in Fig. 1 and a schematic of the machine layout is shown in Fig. 2, illustrating the location of the different power core components and of the maintenance ports. The key machine parameters are summarized in Table I.

The plasma power flow is illustrated in Fig. 3. The core power consists of the alpha power and of any added power to drive the plasma (in the ARIES-CS case, we assume no added power in steady state). The core power is then divided into alpha-particle loss, core radiation, and particle power. For a compact stellarator the alpha loss can be significant, about 5%. The core radiation fraction $f_{rad,core}$ includes bremstrahlung and synchotron radiations. The particle power would then be divided in an edge radiation fraction $f_{rad,edge}$ and the conducted power to the divertor. From the system studies,[7] both $f_{rad,core}$ and $f_{rad,edge}$ are set at 75%. The conducted power reaching the divertor would result in a heat load profile governed by factors such as the divertor inclination and the

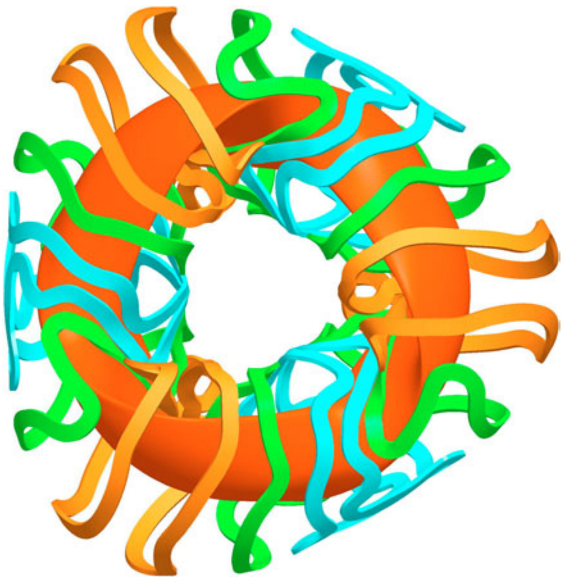

Fig. 1. NCSX-like three-field-period configuration developed for ARIES-CS.

TABLE I

System Parameters of ARIES-CS Three-Field-Period Configuration

Major radius	7.75 m
Minor radius	1.7 m
Aspect ratio	4.5
Minimum coil-plasma distance	1.3 m
β	5%
Number of coils	18
On-axis magnetic field	5.7 T
Maximum magnetic field at coil	15.1 T
Fusion power	2.4 GW
Minimum/average/maximum neutron wall loads	0.32/2.6/5.3 MW/m^2
First-wall surface area	742 m^2
Minimum/average/maximum heat flux on first wall from core radiation	0.20/0.48/0.68 MW/m^2
Minimum/average/maximum overall heat flux on first wall	0.28/0.57/0.76 MW/m^2
Alpha loss	5%

particle intersection footprint on the divertor plates (with a resulting peaking factor $F_{div,peak}$). Edge radiation can be further divided into a component radiating from the vicinity of the divertor region mostly to adjacent first-wall modules, $f_{rad,edge,div}$, and another component radiating to the entire first wall. For the system studies, $f_{rad,edge,div}$ was assumed as 50% with a further assumption that the edge radiation from the divertor region would

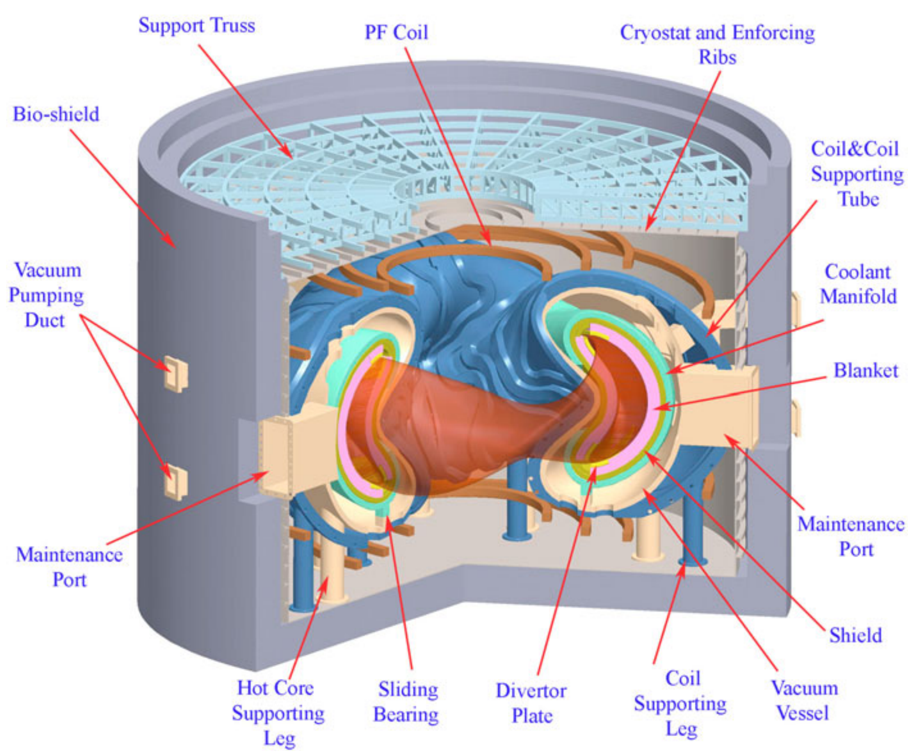

Fig. 2. Schematic of ARIES-CS three-field-period machine layout.

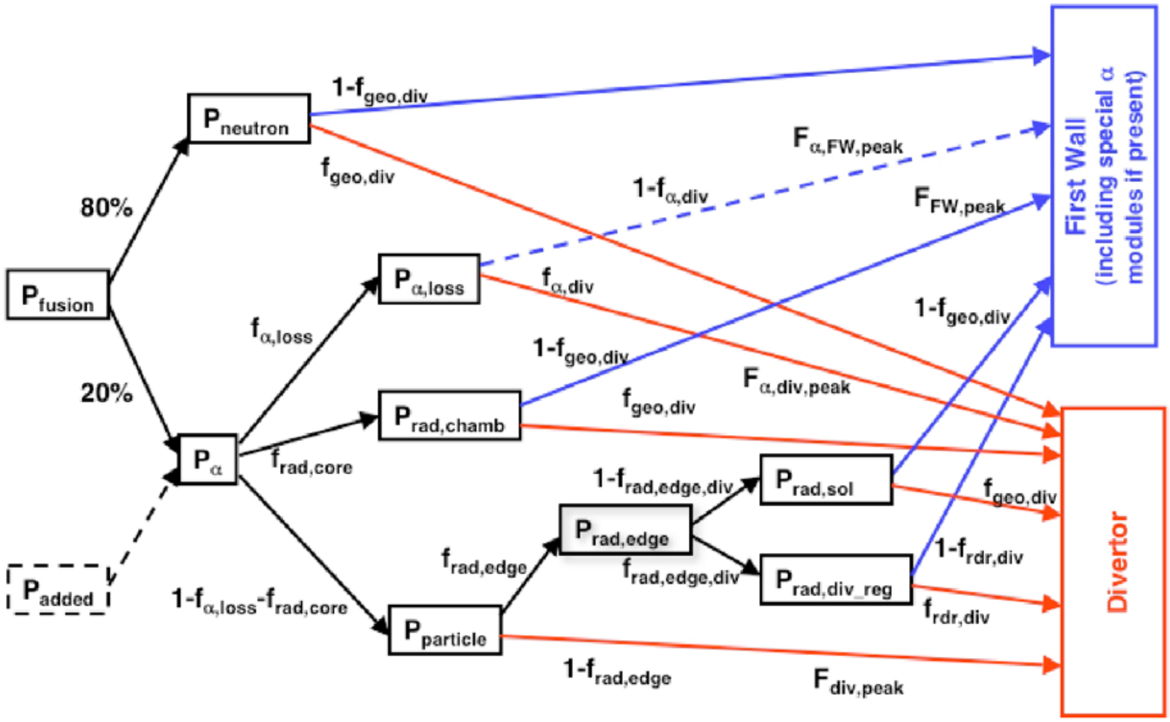

Fig. 3. Plasma power flow.

also be split evenly between radiation to the divertor only and radiation to the rest of the first wall[7] ($f_{rdr,div} = 50\%$).

II.B. Nuclear Analysis

A novel approach based on coupling the CAD model with the neutronics code was developed to model, for the first time ever, the complex stellarator geometry for nuclear assessments to provide detailed neutron wall load and radiation heat flux profiles and to confirm the tritium breeding prediction.[4] From these detailed three-dimensional (3-D) CAD/MCNP calculations, the ratio of maximum-to-average neutron wall load was estimated as 2.04 and the ratio of maximum-to-average first-wall heat flux from core radiation was estimated as 1.43. For the total heat flux on the first wall, including edge radiation and radiation in the vicinity the divertor region, this ratio is about 1.36.

Several additional nuclear-related tasks received considerable attention during the design process. These include blanket breeding/shielding optimization to help minimize the radial build in the local regions where the plasma-to-coil space is constrained and, thus, reduce the overall machine size, as highlighted in Sec. V.A; optimization of radial build definition and of in-vessel component design to satisfy the top-level safety and operational requirements; assessment of streaming of neutrons through the helium access tubes and pipes to provide design guidance; careful selection of nuclear and engineering parameters to produce an economic optimum; and adoption of overarching safety constraints to deliver a safe and reliable power plant. These are discussed in more detail in Ref. 4.

III. MAINTENANCE SCHEME

Two major maintenance schemes were considered as part of the ARIES-CS study: field-period–based maintenance and port-based maintenance. The first approach comprises removal of complete field-period sectors. In this approach, the vacuum vessel is external to the coils and has to be opened first. Next, the field-period sectors are translated radially outward a short distance and the blanket and divertor modules are then removed toroidally in sequential order. The convoluted geometry of the first wall, blanket, and shield required a 3-D analysis to verify that the blanket modules could be extracted toroidally without interference with the outer hot shield. More details can be found in Ref. 5. Under this approach, a very large and massive field-period sector of the power core (approximately 4×10^6 kg for the three-field-period configuration, and even more if a two-period configuration were to be chosen) must be moved, the superconducting coils have to be deenergized and warmed, and all lifetime components have to be disconnected and realigned upon reinsertion. Such considerations weighed against this approach, and port-based maintenance was selected as the reference maintenance scheme.

Port-based maintenance involves the replacement of blanket modules using an articulated boom through a number of designated ports. In this approach, the vacuum vessel is internal to the coils and serves as an additional shield for the protection of the coils from neutron and gamma irradiation, as illustrated in Fig. 2. During maintenance, no disassembling and rewelding of the vacuum vessel are required and the modular coils after being deenergized can be kept at cryogenic temperatures. For the NCSX-like three-field-period configuration, one main maintenance port and one auxiliary maintenance port are assumed per field period. A fixed transfer chamber permanently attached to the bioshield is used in transferring out the modules being replaced and transferring in the new modules. The size of the transfer chamber is large enough to accommodate the maintenance equipment and removed and stored power core components. The size of the main port [1.85 m (toroidal) × 3.85 m (poloidal)] and the boom load capacity and its required reach limit the weight (~5000 kg), geometry, and size of the replaceable units (first wall and blanket, divertor, and a few elements of the shield). An electron cyclotron heating (ECH) port is included in each field period, as illustrated in Fig. 4; it is also used as the auxiliary maintenance port once the ECH launcher assembly is removed. Details on the maintenance procedures can be found in Ref. 8.

IV. POWER CORE INTEGRATION

The power core configuration is illustrated in Figs. 2 and 4; it includes a breeding zone subdivided into blanket modules, shield and manifold regions, a vacuum vessel internal to the coil structure, and maintenance ports. A key aim in evolving the design integration of the power core was to minimize thermal stresses. The hot core region (including shield and manifold at ~450°C) forms part of a strong skeleton shell (continuous poloidally, divided toroidally in sectors). Each skeleton shell sector rests on sliding bearings (possibly made of tungsten carbide) at the bottom of the cooler vacuum vessel (~200°C) and can freely expand relative to the latter. The blanket modules are mechanically attached to the skeleton shell and can also float with it relative to the vacuum vessel. Bellows are used between the vacuum vessel and the coolant access pipes at the penetrations. These bellows provide a seal between the vacuum vessel and cryostat atmospheres and see only minimal pressure difference. In addition, as explained in the blanket section, concentric coolant access pipes are used for both He and

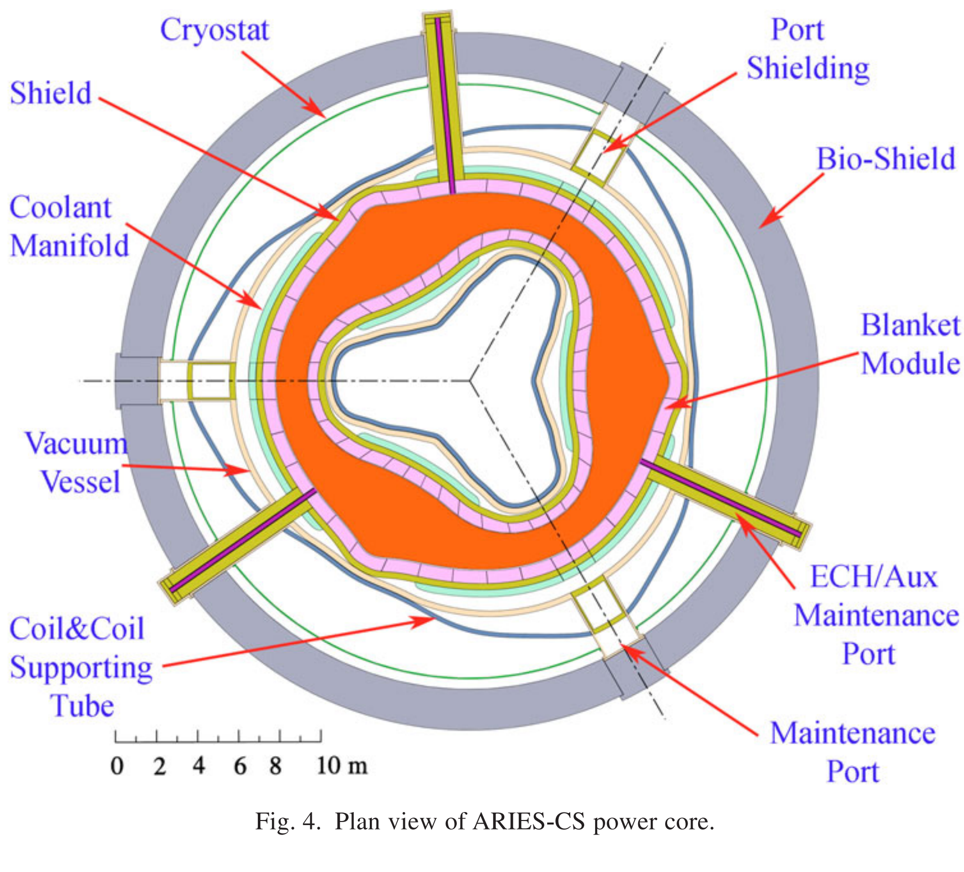

Fig. 4. Plan view of ARIES-CS power core.

Pb-17Li, with the return He in the annular channel (at ~460°C) in the first case and the inlet Pb-17Li in the annular channel (at ~460°C) in the second case to maintain near-uniform temperature in the skeleton shell. Furthermore, temperature variations in the blanket module are minimized by cooling the steel structure with He (with a temperature rise <100°C).

Provisions for cutting and rewelding of the vacuum vessel have to be made only for the very unlikely case that coils have to be replaced or the vacuum vessel itself fails. Considering the nonuniform shape and size of the modular coils, the vacuum vessel for the ARIES-CS with three field periods is assembled from six sectors. The assembly welds are arranged at the largest cross section (at 0 deg) and the smallest cross section (at 60 deg). This allows sliding of the vacuum vessel sector into the coils of a field period in the toroidal direction.

The coil structure, described in more detail in Sec. VII, consists of three thick-walled supporting tubes (one per field period) bolted together to form a strong and continuing structural ring, as illustrated in Fig. 5. The six modular coils of each field period are wound internally into grooves in the coil structure. The weight of the cryogenic coils and coil structure is transferred to the bottom foundation via three long legs per field period with high thermal resistance in order to keep the heat ingress into the cold system to tolerable limits. A number of penetrations through the coil structure are required, as shown in Fig. 5. These include the main maintenance ports, the ECH/auxiliary maintenance ports, and accesses for coolant lines, pumping, and supporting structure. More details can be found in Ref. 8.

The entire coil system is enclosed in a common cryostat (as illustrated in Fig. 2) since no disassembly is necessary for the blanket exchange. Thermal insulation between the cold coil structure and the warm vacuum vessel has to be provided. The cryostat consists of a cylindrical section with reinforcing ribs and top and bottom sections. It can serve as a vacuum environment to limit thermal loads to the superconducting magnet system and at the same time as a second containment for the tritium in the vacuum vessel.

V. BLANKET AND ANCILLARY EQUIPMENT

A number of different blanket concepts were considered during the early phase of the study, including the following[3]:

1. self-cooled flibe blanket with advanced ferritic steel
2. self-cooled Pb-17Li blanket with SiC_f/SiC composite as structural material

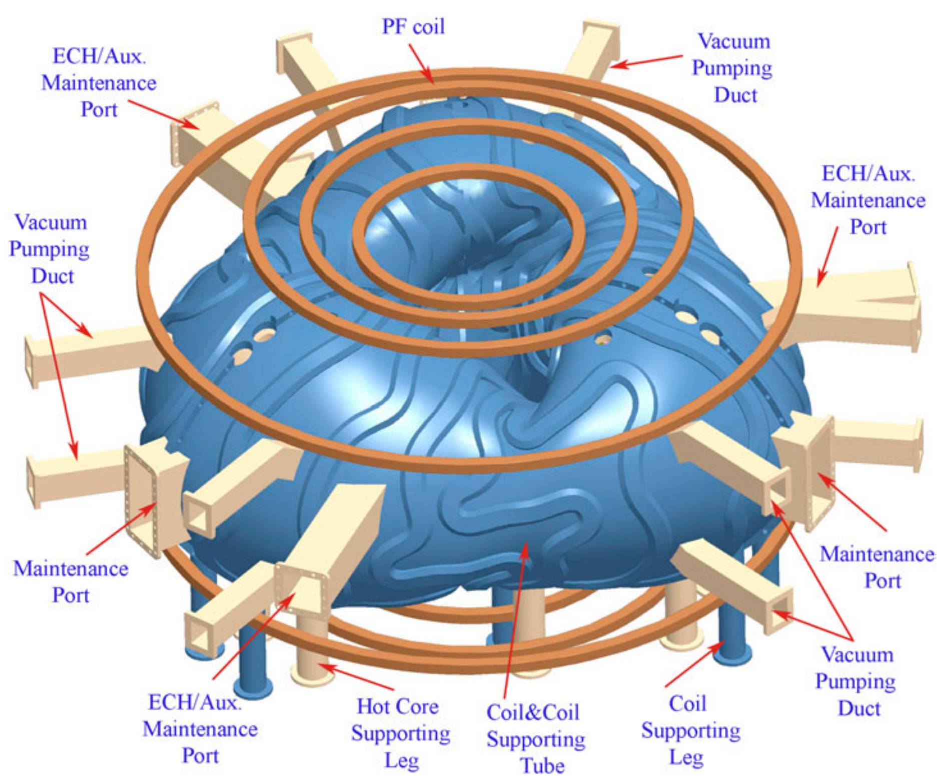

Fig. 5. Schematic of coil structure including the different penetrations.

3. dual-coolant blanket concept with He-cooled steel structure and self-cooled liquid metal (Li or Pb-17Li breeding zone)
4. helium-cooled ceramic breeder blanket with ferritic steel structure.

Different quantitative and qualitative considerations led to the choice of a dual-coolant concept with self-cooled Pb-17Li zones and He-cooled ferritic steel structure as reference blanket option.[3] A self-cooled Pb-17Li blanket with SiC_f/SiC composite as structural material, based on the ARIES-AT blanket design,[9] is maintained as a backup option. It allows for high coolant temperature (~1000°C to 1100°C) and high performance but carries a higher development risk (mostly due to SiC_f/SiC material development).

In the final phase of the study, the engineering effort concentrated on the reference dual-coolant blanket design, which is the focus of this section. Some modest design and analysis effort was spent on the advanced option, mostly to adapt the ARIES-AT blanket design to the CS case to estimate the corresponding radial build and neutronics parameters[4] and the cycle efficiency (~55 to ~60% for neutron wall loads of 5 to 2 MW/m^2). The results were then used as input for one of the trade-off studies to assess at the system level the overall performance of a CS power plant with such an advanced blanket.[7]

V.A. Blanket Modules

A key engineering parameter for a compact stellarator is the minimum plasma-to-midcoil distance that dictates the minimum reactor size. It should accommodate the scrape-off layer, first wall, blanket, shield, manifolds, vacuum vessel, assembly gaps, coils case, and half of the winding pack. An innovative approach was developed to downsize the blanket in these space-constrained regions and utilize a highly efficient tungsten carbide (WC)-based shield. The special modules in these regions utilize a nonuniform blanket and a WC shield, optimized to provide shielding comparable to a regular breeding module but with a much reduced module radial thickness, as shown in Fig. 6. From the figure, the typical plasma-to-midcoil radial thickness for a regular breeding module is about 1.79 m but is only 1.31 m for an optimized module with a reduced breeding zone but with similar shielding properties.[4] The final design of the blanket modules in those regions with minimum plasma-to-midcoil distances (for example, in the inboard region of the toroidal location shown in Fig. 2) consists of a tapered breeding region with a thickness ranging from 25 cm at the minimum space location to 54 cm in regions where more space is available. The total tritium breeding including all modules is ~1.1. Details of the shielding and breeding study can be found in Ref. 4.

V.B. Dual-Coolant Blanket Module Layout

The dual-coolant concept utilizes He to cool the ferritic steel structure (including the first wall) and slowly flowing Pb-17Li as self-cooled breeder in the inner channels, which can be operated at a higher temperature than the structural blanket walls to maximize the power cycle

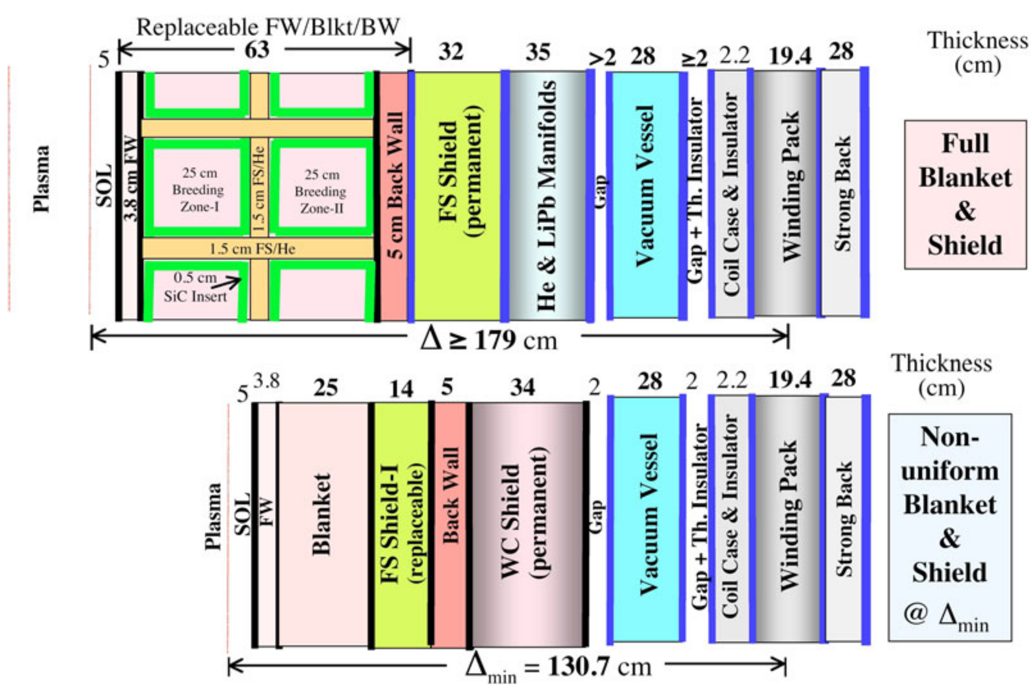

Fig. 6. Optimized blanket and shield radial build.

efficiency. Use of He coolant is compatible with the coolant strategy for the reactor since He is also used to cool the divertor. The use of this coolant for the first wall/structure in dual-coolant blankets also facilitates the preheating of the blankets before the liquid breeder is filled in, serves as guard heating in case the liquid breeder cannot be circulated, and provides independent and redundant decay heat removal in case the liquid metal loop is not operational. In addition, cooling the first-wall region of the blanket (where the heat load is highest) with helium (instead of a liquid metal) avoids the need for an electrically insulating coating in this high-velocity region to prevent the large magnetohydrodynamic (MHD) pressure drop associated with liquid metal flow.

Such a dual-coolant concept was originally developed as part of the ARIES-ST study[10] and then at FZK in Germany.[11] It is also now being considered as a U.S. test module candidate for ITER (Ref. 12). A modular concept was adapted for the ARIES-CS geometry with a particular focus on developing a more efficient coolant routing configuration and on optimizing the design performance of the blanket coupled to a Brayton power cycle.[13]

The blanket module concept is illustrated in Figs. 7 and 8, with both He and Pb-17Li fed to the back of the blanket module through concentric pipes. The helium coolant (assumed at 10 MPa) is routed to first cool the first wall in a single pass with an alternating toroidal flow configuration to create a more uniform temperature (and reduce thermal stresses); it is then routed in a combination of series and parallel flow to cool the other structural walls shown in Fig. 8. The Pb-17Li flows slowly (~10 cm/s or less) in the large inner channels in a two-pass poloidal configuration, as illustrated in Fig. 9. The Pb-17Li channels are lined with a SiC flow channel insert (FCI) (with no structural function since the thin Pb-17Li layer and the bulk Pb-17Li are pressure balanced through a thin slot in the SiC), as illustrated in Fig. 10. This FCI plays a key thermal insulation function to allow high-temperature (~700°C) Pb-17Li in the channel while the Pb-17Li/FS interface temperature is maintained below its compatibility limit. There are 198 blanket modules with typical first-wall areas between 3 and 4 m^2. The major blanket parameters are summarized in Table II.

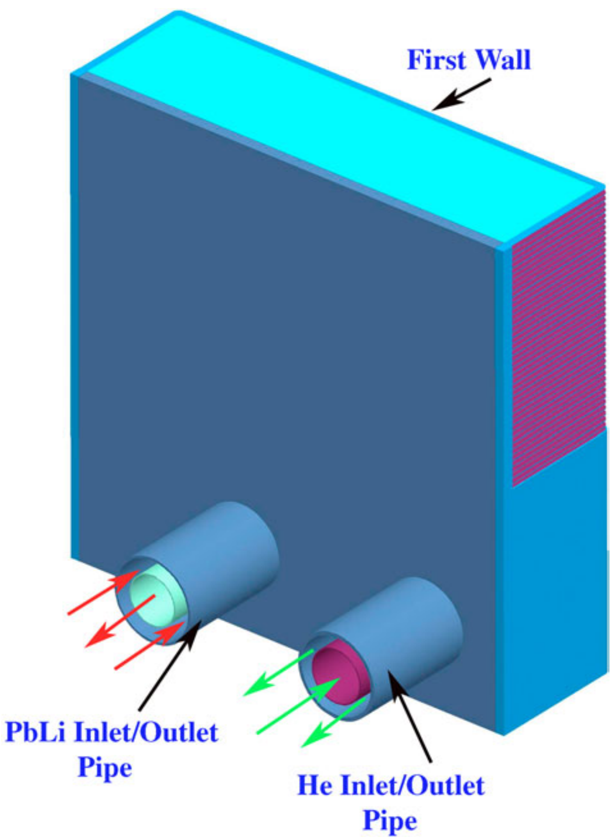

Fig. 7. ARIES-CS dual-coolant blanket module.

V.C. Module Attachment

The blanket modules (Fig. 11) have been designed so that the two concentric coolant pipes at the back of the module (servicing the He coolant and Pb-17Li coolant, respectively, as shown in Fig. 7) can be cut from the outside following removal of shield pieces protecting the rewelding area from neutron streaming (less than ~1 at. ppm He). This requires that an adjacent module be first removed starting with the port module to provide access for the maintenance operation. For the Pb-17Li concentric pipe, the inlet flow is routed through the annular channel and the outlet flow through the inner channel. For both He and Pb-17Li cooling pipes, only the outer tube needs to be cut and rewelded, the inner tube being attached by a slip joint (as is the SiC insert also in the Pb-17Li case).

The He and Pb-17Li coolant piping also serves to mechanically attach the blanket module to the hot shield and coolant manifold/structure, which are lifetime components. These are in turn vertically supported through a sliding bearing on the vacuum vessel. In addition, a mechanical attachment is also needed to help stabilize each blanket module and support it during disassembly. For example, a system with four bolts (~2 to 3 cm in diameter) coming from the shield and supporting the module at its sides could be used. Nuts with an internal hexagonal opening for the tool placement would then be screwed in from a small front access hole but at the level of the back plate; the nut dimensions would have to be optimized to provide the required shielding for the bolt and lifetime component at the back while allowing for acceptable cooling by local radiation to the module (initial estimates indicate that a temperature of ~500°C for the bolt and ~600°C for the nut would be required). Shielding optimization studies have not been performed to this level of detail for this concept, with the understanding

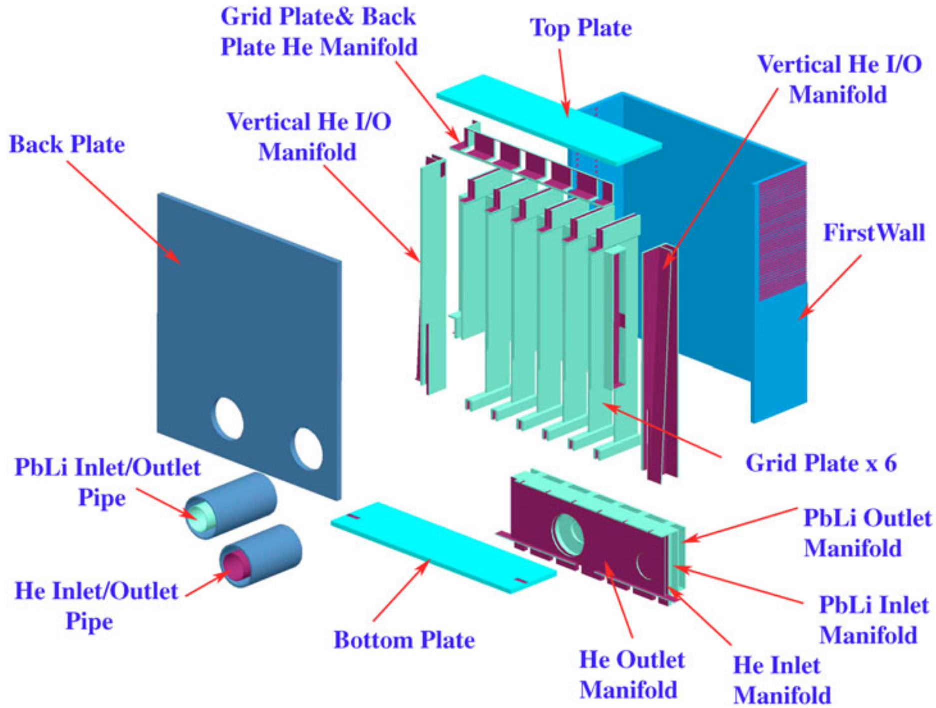

Fig. 8. Exploded view of the ARIES-CS blanket module

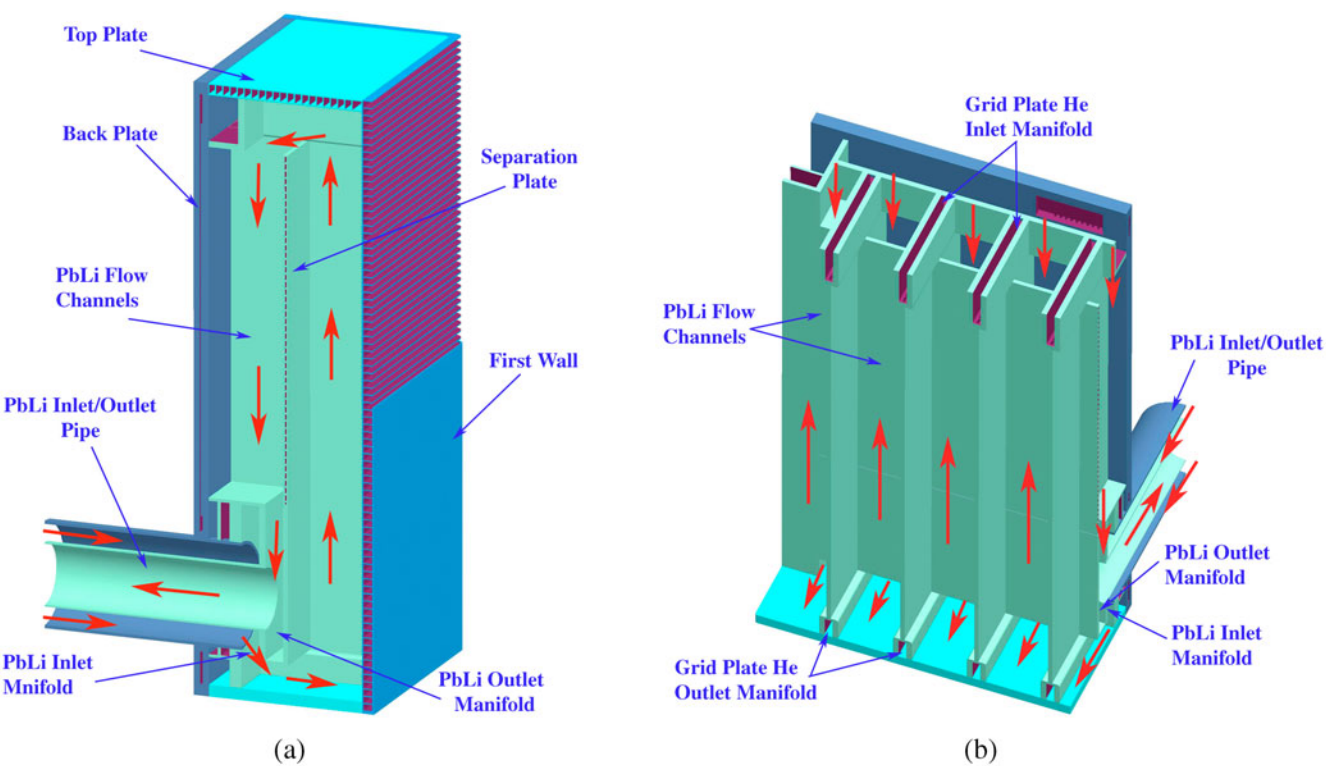

Fig. 9. Details of Pb-17Li flow scheme in ARIES-CS blanket module.

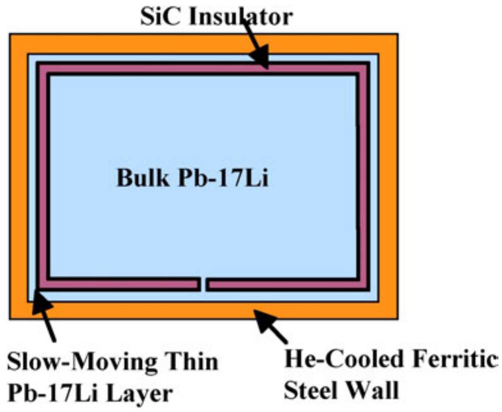

Fig. 10. Schematic of SiC insert in Pb-17Li channels.

TABLE II

Summary of ARIES-CS Blanket Parameters

Typical module dimensions	$\sim$4 m^2 × 0.62 m
Number of modules	198
Tritium breeding ratio	1.1
Neutron energy multiplication factor	1.16
Fusion thermal power in blanket	2496 MW
Pb-17Li inlet/outlet temperatures	451/738°C
Pb-17Li inlet pressure	1 MPa
Typical inner channel dimensions	0.26 m × 0.24 m
Effective SiC insulator region conductivity	200 W/m^2·K
Average Pb-17Li velocity in inner channel	$\sim$0.04 m/s
Fusion thermal power removed by Pb-17Li (including $\sim$111 MW reduction due to conducted power to He)	1444 MW
Pb-17Li total mass flow rate	26 860 kg/s
Pb-17Li pressure drop	$\sim$ 0.1 to 1 kPa
Pb-17Li pumping power	$\sim$ 1 to 10 kW
Maximum Pb-17Li/FS temperature	468°C
He inlet/first-wall outlet/module outlet temperatures	386/430/456°C
He inlet pressure	10 MPa
Typical first-wall channel dimensions (poloidal × radial)	2 cm × 3 cm
He velocity in first-wall channel	52 m/s
He heat transfer coefficient in first-wall channel	8825 W/m^2·K
Total blanket + heat exchanger He pressure drop	0.30 MPa
Thermal power removed by He (including 141 MW of friction power + 111 MW of conducted power from Pb-17Li)	1192 MW
Total mass flow rate of blanket He	3261 kg/s
Blanket He pumping power	156 MW
Maximum volumetric heat generation in first wall	$\sim$44 MW/m^3
Maximum local ODS FS temperature in first wall	$\sim$642°C
Maximum local RAFS temperature in first wall	$\sim$546°C
Maximum primary stress blanket module	$\sim$125 MPa
Maximum primary + secondary stress in ODS FS in first wall (2-D under plane stress/plane strain assumptions)	$\sim$472/654 MPa
Maximum primary + secondary stress in RAFS in first wall (2-D under plane stress/plane strain assumptions)	$\sim$326/390 MPa

that the nut design can only be finalized based on such studies.

More details of the blanket attachment, removal, and maintenance can be found in Ref. 8.

V.D. Material Constraints

Reduced activation ferritic steel (modified F82H) is utilized as the main structural material, with a maximum temperature limit of $\sim$550°C based on thermal creep considerations.[14] In regions with higher temperature, it is possible to use oxide dispersion–strengthened ferritic steel (ODS FS) with a much higher temperature limit ($\sim$700°C) (Ref. 14). However, joining is an issue, and it is desirable to limit the use of ODS FS as much as possible. A more constraining limit is the interface temperature between the thin Pb-17Li layer and the FS wall (see Fig. 10). A compatibility limit of 450°C (Ref. 14) has been assumed for quite some time for FS/Pb-17Li based on an old data set and on corrosion criteria of ambiguous relevance to the blanket. However, it was decided in anticipation of updated results not to overly penalize the blanket design and to assume that the interface temperature between FS and Pb-17Li could be up to $\sim$500°C in limited regions.

V.E. Ancillary Equipment and Power Cycle

The blanket is coupled to a Brayton cycle through a heat exchanger (HX), where both coolants (He and Pb-17Li) transfer their energy to the cycle working fluid (He), as illustrated in Fig. 12. For the Pb-17Li, use of a SiC insert in the return inner channel of the concentric coolant access pipes would serve as an insulator to minimize the heat flow from the hot side to the cold side as well as to maintain the Pb-17Li/ferritic steel interface temperature within its allowable limits, as explained in Sec. V.F. Thus, ferritic steel can be considered for the coolant piping; however, the heat exchanger tubes would have to be made of higher-temperature material (most probably a refractory alloy, such as niobium or tantalum). Use of niobium or tantalum alloys, which have a high tritium permeability, is also considered for a vacuum permeator tritium recovery system from the hot Pb-17Li, which would then be routed to the heat exchanger with a minimal remaining tritium concentration (thus minimizing the tritium permeation there).[15] The vacuum permeator extraction consists of a bank of small-diameter, thin-walled, niobium alloy tubes through which the entire Pb-17Li primary coolant flows at 5 m/s. At the outside of the tube bundle is a high-vacuum region. Because the Pb-17Li flow is turbulent at this velocity, the mass

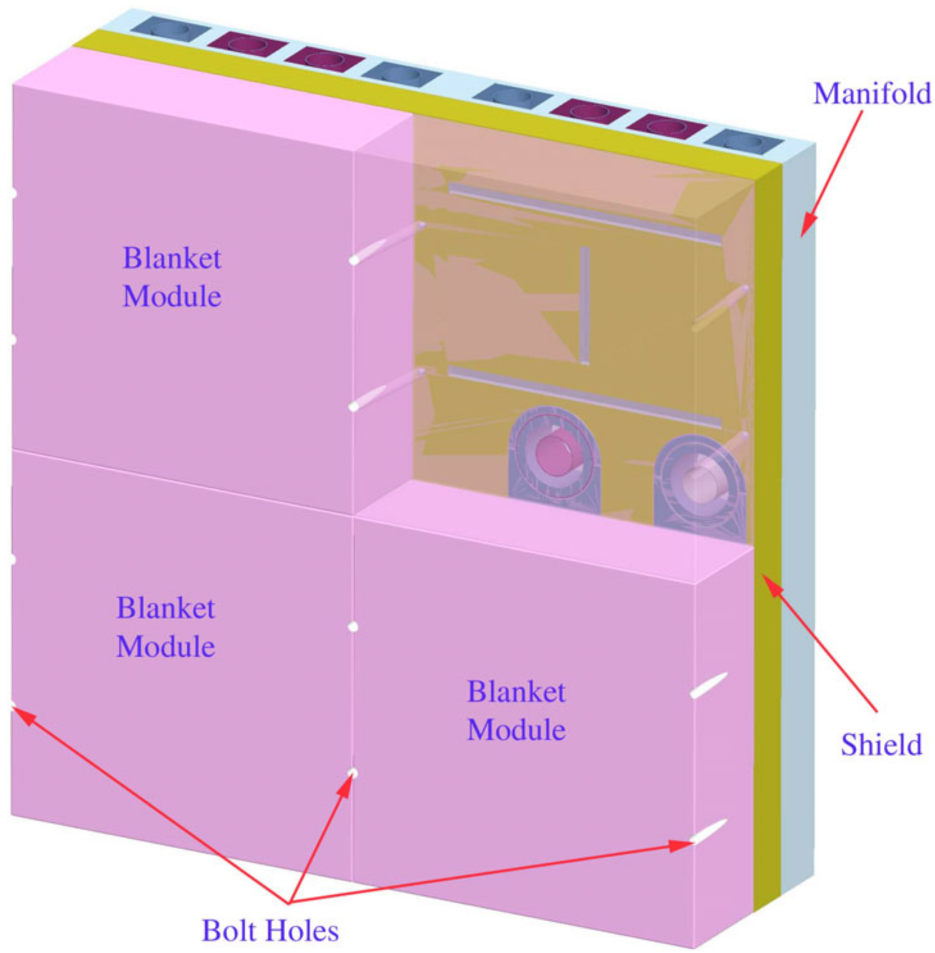

Fig. 11. Schematic of four blanket modules attached to the shield including a transparent view of one of the modules.

transport of tritium to the Pb-17Li/tube interface is greatly enhanced over that of ordinary diffusion. Once at this surface, the tritium readily diffuses through the niobium tube walls and into the vacuum region, where it is pumped away in the elemental form to the tritium processing plant. The resulting partial pressure of tritium above the Pb-17Li breeder entering the heat exchanger is low enough (~0.7 Pa) to maintain ARIES-CS heat exchanger tritium inventories below 200 g (Ref. 16).

The Brayton cycle considered for this study includes three compression stages and a single expansion stage,[15] as illustrated in Fig. 13. Its assumed parameters are summarized in Table III.

V.F. Thermal-Hydraulic Analysis

Detailed thermal-hydraulic analyses were performed for the dual-coolant blanket coupled to the Brayton cycle with the goal of optimizing the parameters for best performance. The analysis included calculating the maximum net cycle thermal efficiency for different neutron wall loads (and corresponding plasma heat fluxes) for a given set of constraints, including a maximum Pb-17Li/FS interface temperature <500°C. For a radially averaged FS temperature at the first wall <550°C [for reduced activation ferritic steel (RAFS)], the maximum wall load was limited to ~3 MW/m^2, which overly limits the minimum major radius of the machine for a given configuration. Use of ODS FS for the first wall as an ~3 mm layer bonded to an ~1 mm RAFS sublayer allows for higher operating temperature and a higher neutron wall load.

It is desirable to minimize the thermal conductance of the SiC flow channel insert to allow for a high Pb-17Li outlet temperature while accommodating the 500°C Pb-17Li/FS maximum interface temperature limit assumed here. The overall thermal conductance of this insulation region was set as 200 W/m$^2\cdot$K under the assumption that it could be credibly achieved (e.g., through a 5-mm-thick layer of porous SiC with a conductivity of 1 W/m$\cdot$K assuming a uniform bulk Pb-17Li temperature). Although the MHD pressure drop of Pb-17Li in bare inner

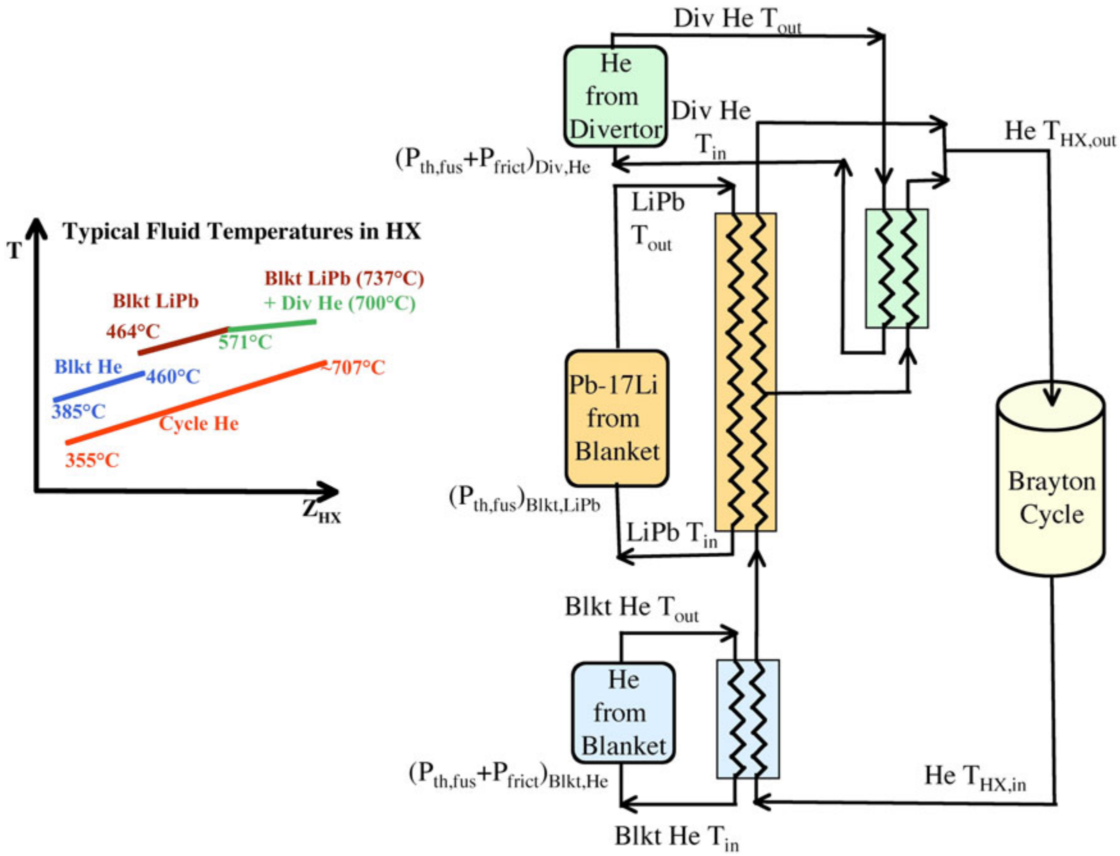

Fig. 12. Heat exchangers coupling blanket and divertor coolants to the power cycle.

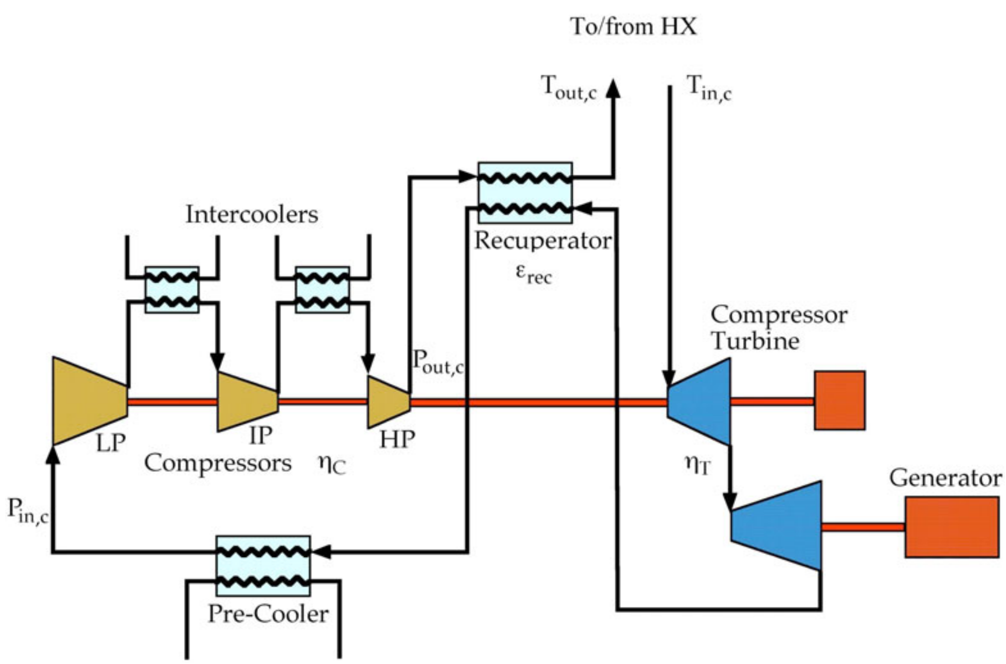

Fig. 13. Schematic of Brayton cycle considered in this study ($T_{in,c}$ and $T_{out,c}$ = cycle He temperatures from and to HX; $P_{in,c}$ and $P_{out,c}$ = cycle He pressures in and out of three-stage compression with LP = low pressure, IP = intermediate pressure, and HP = high pressure).

TABLE III

Summary of Brayton Power Cycle Parameters

Number of compression stages	3
Number of expansion stages	1
HX temperature difference between hot and cold legs	30°C
Compressor efficiency, η_C	0.89
Turbine efficiency, η_T	0.93
Recuperator effectiveness, ε_{rec}	0.95
Cycle maximum He pressure, $P_{out,c}$	15 MPa
Total compression ratio, $(P_{out,c}/P_{in,c})$	3.5
Fractional cycle He pressure drop, $(\Delta P_{fric,c}/P_{out,c})$	0.045
Cycle lowest He temperature	35°C
Gross efficiency	0.43
Net efficiency (+ contribution of blanket and divertor He friction thermal power − pumping power)	0.39

channels could be acceptable (~0.1 to 0.5 MPa), the uncertainty linked with turning flows could increase the total pressure to an unacceptable value. Thus, the SiC FCI also plays an important electrical insulation function, which was included in the flow calculations.

A numerical study was performed of the MHD effect on the Pb-17Li flow distribution and behavior in the inner channels as well as in the gap between the SiC FCI and the FS wall. The analysis considered the electrical coupling between six adjacent poloidal channels with an assumed Hartmann (Ha) number of 5000 and investigated the influence of various parameters such as the electric conductivity of the SiC insert and the orientation of the magnetic field. Details of the study are described in Ref. 17. Some of the key findings are summarized as follows:

1. The pressure equalization slot in the FCI should be placed on the side perpendicular to the magnetic field (Ha wall) to avoid strong local reverse flow.

2. There is a weak electric coupling among the channels, shown by the fact that no current crosses the separation plate.

3. Due to the presence of the FCIs, the side walls behave almost as an insulating material and their properties have limited effects on the velocity distribution in the inner channels.

4. The presence of increased velocity along the side walls, due to the imperfect insulation of the Hartmann walls, may be regarded as a positive feature since it can reduce tritium permeation to the helium cooling channels.

5. Further reduction of the electrical conductivity of the FCI below about 100 $\Omega^{-1}\,m^{-1}$ has minimal effect on the velocity distribution, indicating good insulation quality at this value.

The overall thermal-hydraulic optimization was done by considering the efficiency of the Brayton cycle for an example 1000 MW(electric) case. The friction power from the He flow in the blanket and divertor was added to the fusion thermal power for these calculations, and the thermal-hydraulic parameters of the blanket were optimized to maximize the net cycle efficiency (based on the gross electrical power minus the pumping power) for the given set of material constraints and cycle parameters. The results are summarized in Tables II and III for the reference case, with gross and net cycle efficiencies of ~0.43 and ~0.39, respectively. Increasing the machine size for the same fusion power (i.e., reducing the maximum wall load and heat flux on the wall) would help increase the coolant temperatures and corresponding efficiency for the given material constraints. This is illustrated in Fig. 14, which shows the gross cycle efficiency as a function of the first-wall surface area for a given fusion power. The corresponding maximum neutron wall load (also shown on the upper x-axis of Fig. 14) and maximum surface or plasma heat flux (based on the assumptions described in Sec. II.A) are shown in Fig. 15. In all cases the maximum RAFS temperature is <550°C and the FS/Pb-17Li interface temperature is <500°C; the other parameters are consistent with those in Tables II and III. The effect of changing the power density is more apparent on the blanket He pumping power, which increases appreciably with decreasing first-wall surface area (or increasing neutron wall load), as also shown in Fig. 14. The power core net cycle efficiency based on the gross

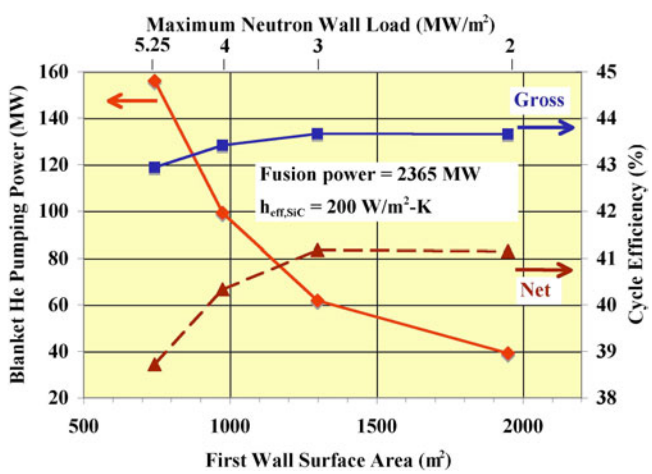

Fig. 14. Blanket He pumping power as a function of the first-wall surface area for a fusion power of 2.36 GW. Also shown on the second y-axis are the gross and net cycle efficiencies as a function of the first-wall surface area. The corresponding maximum neutron wall load is shown on the upper x-axis for information.

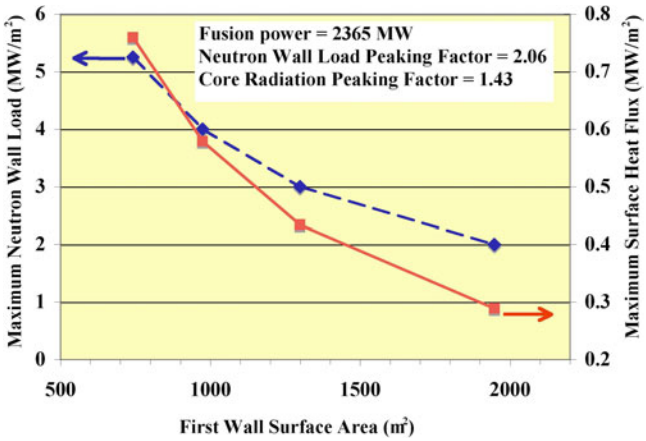

Fig. 15. Maximum neutron wall load and maximum first-wall surface heat flux as a function of the first-wall surface area for a fusion power of 2.36 GW and maximum-to-average peaking factors of 2.06 and 1.43 for the neutron wall load and core radiation heat flux, respectively.[4]

electrical power minus the coolant pumping power also decreases markedly as the power density is increased, as shown in Fig. 14.

The effect of changing the effective conductance ($h_{eff,\mathrm{SiC}}$) of the SiC insert region illustrated in Fig. 10 was also investigated. In these calculations $h_{eff,\mathrm{SiC}}$ was assumed to include the thermal conductivity of the SiC region itself as well as the effective heat transfer coefficients at the Pb-17Li/SiC interface on each side of the insert. An accurate analysis would require detailed 3-D MHD analysis of the Pb-17Li velocity profile, volumetric heat generation, and thermal behavior, which is beyond the scope of this study. Here, the calculations were intended to provide only a rough estimate of the thermal behavior of the SiC FCI in the blanket; for simplicity, the overall heat transfer between the higher-temperature Pb-17Li and the lower-temperature He was based on their average temperatures, whereas the maximum Pb-17Li/FS interface temperature was found to occur in the separation plate region with on one side Pb-17Li assumed at its outlet temperature and on the other side He at its outlet temperature.

A lower conductance was found to reduce the heat conducted from the higher-temperature Pb-17Li to the lower-temperature He in the blanket, thereby reducing the blanket He mass flow rate and pumping power. Increasing the effective conductance does not affect the gross cycle efficiency but has a small effect on the net cycle efficiency due to the added power that needs to be removed by the blanket He and the corresponding pumping power. For example, increasing $h_{eff,\mathrm{SiC}}$ from 20 to 800 $W/m^2 \cdot K$ would result in the He pumping power increasing from ~150 to ~165 MW and the net cycle efficiency decreasing from ~0.389 to ~0.384%. Results of the analysis also indicated that increasing $h_{eff,\mathrm{SiC}}$ does not affect the maximum RAFS temperature (which was ~550°C for all the cases) but has an effect on the maximum FS/Pb-17Li interface temperature, which increases from ~454 to ~480°C (still within the assumed ~500°C limit) as $h_{eff,\mathrm{SiC}}$ is increased from 20 to 800 $W/m^2 \cdot K$. These results are approximate since they are based on simple assumptions and would need to be confirmed by more detailed analysis; however, they are very encouraging since they show that the performance parameters tend to be only slightly affected by changes in $h_{eff,\mathrm{SiC}}$.

Overall, the gain in power core net efficiency as the power density is reduced has to be judged at a system level against other factors such as the increase in cost linked with a larger machine. These trade-off evaluations were performed as part of the overall system optimization studies.[7] Results similar to those shown in Fig. 14 were provided as input for these system runs.

V.G. Stress Analysis

The high neutron wall load and heat flux on the first wall make it quite challenging to maintain the blanket module design within both allowable temperature and stress limits. ANSYS analyses were carried out on a two-dimensional (2-D) section of the module to determine the temperature and stress distributions in the blanket module including a first wall with a 3-mm ODS FS layer over a 1-mm RAFS layer. The analysis was carried out for a neutron wall load of 5.25 MW/m^2 (with a corresponding maximum volumetric heat generation of 44 MW/m^3 in the first wall), a surface or plasma heat flux (q'') of 0.76 MW/m^2, and a reference temperature based on the He inlet temperature (~386°C).

The resulting temperature distribution is shown in Fig. 16a for the whole cross section and in Fig. 16b in a magnified view of the first-wall region. For the blanket parameters, shown in Table II, the maximum temperatures for ODS FS and RAFS are acceptable, ~642 and ~546°C, respectively. The stress calculations were performed under both plane stress and plane strain assumptions, which provide lower and upper bound values for the stress based on assumption of no stress in the third direction and no displacement in the third direction, respectively. The maximum combined stresses (primary + secondary) under these conditions are ~472 and ~654 MPa, respectively, for ODS FS and ~326 and ~390 MPa, respectively, for RAFS. An example stress distribution for the plane strain case is shown in Fig. 17a for the whole cross section and in Fig. 17b in a magnified view of the first-wall region.

Here, the assumed stress design criteria are to limit the primary stress to the allowable primary membrane stress intensity S_m and the primary + secondary stresses to $3S_m$. S_m is a temperature- and fluence-dependent allowable stress intensity defined as the least of different

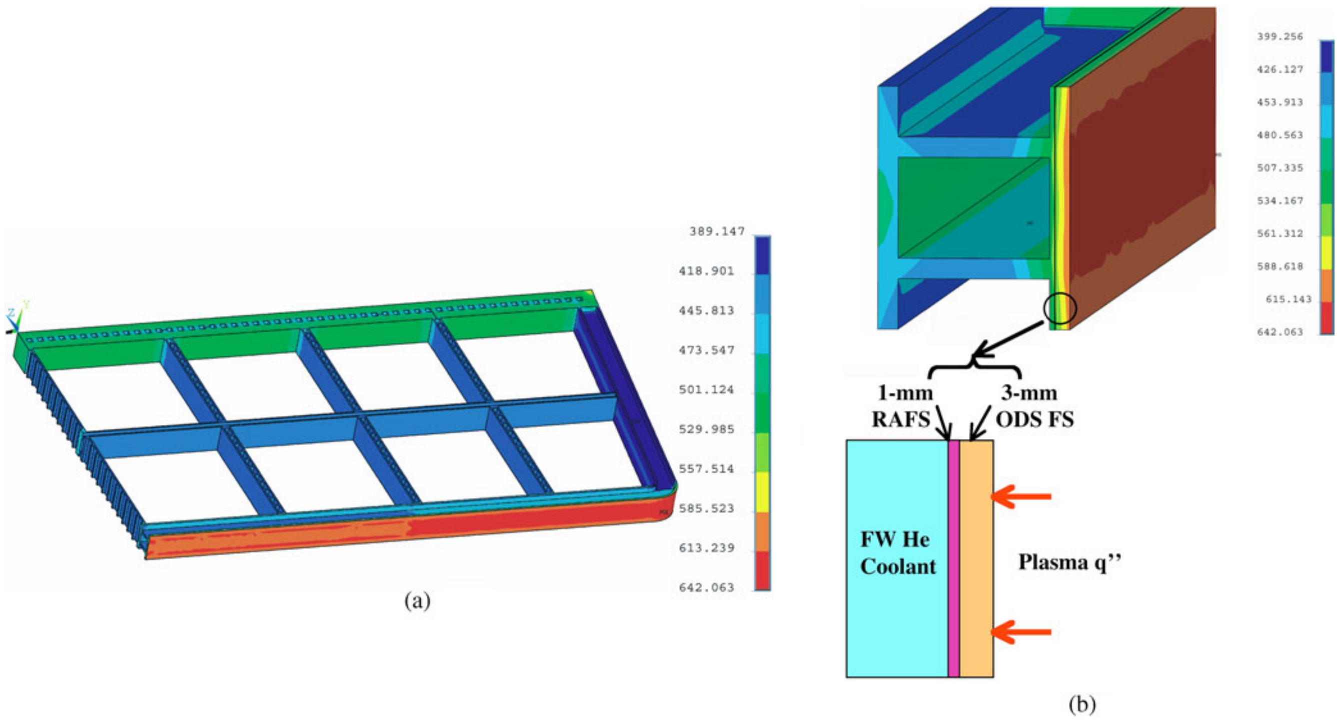

Fig. 16. Temperature distributions (°C) in (a) blanket module cross section and (b) magnified view of first-wall region.

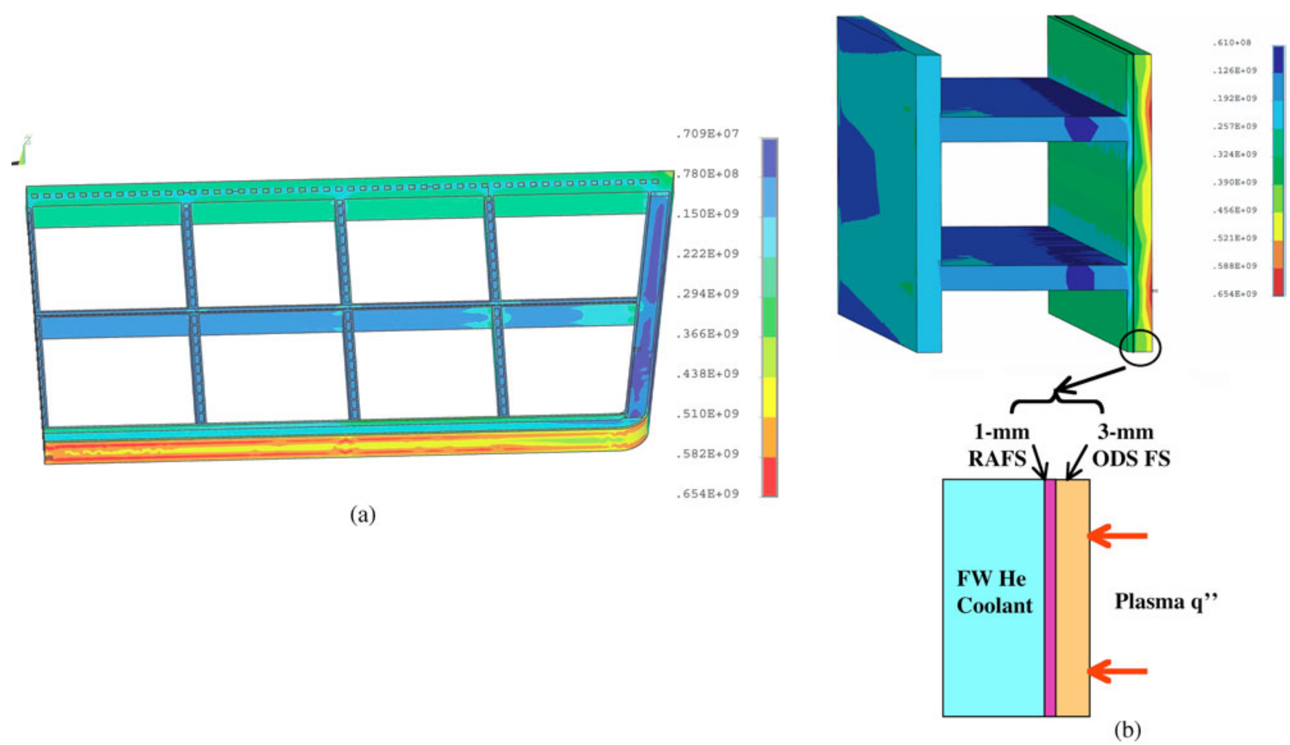

Fig. 17. Primary + secondary stress distributions (Pa) in (a) overall module and (b) magnified view of first-wall region based on plane strain assumption.

TABLE IV

Estimated S_m Values for RAFS and ODS FS*

Alloy	T (°C)	S_m (MPa)	$3S_m$ (MPa)
F-82H	500	133	399
	550	118	354
	600	101	303
ODS LAF-3	500	268	804
	650	133	399
	700	111	333
ODS 12YWT	500	≤500	≤1500
	550	≤460	≤1380
	600	≤420	≤1260
	650	≤220	≤660
	700	≤210	≤630
	750	≤170	≤510
	800	≤155	≤465

*Reference 19.

quantities (such as two-thirds of the yield stress and one-third of the ultimate tensile strength).[18] Table IV shows the S_m values as a function of temperature for F-82H (as representative of an RAFS) and for two representative ODS FS: ODS LAF-3 and ODS 12YWT (obtained from Ref. 19).

The maximum primary stress was found to be ~125 MPa, close to the S_m limit for RAFS (~118 MPa at 550°C) and within the S_m limit for both ODS FS at 654°C. However, the maximum primary + secondary stress based on the plane strain assumption is above the limit for RAFS (~390 MPa compared to $3S_m \sim 354$ MPa at 550°C) and, in the case of ODS FS, can only be possibly met with the ODS 12YWT FS (~654 MPa compared to $3S_m \leq 660$ MPa at 650°C). It is possible that the total stress would be lower than the plane strain numbers and within the design constraints; however, this shows that the design has been pushed to its limit and that it would be beneficial to relax the heat load to provide more margin (for example, by increasing the reactor size).

VI. PLASMA-FACING COMPONENTS

A concerted effort on the divertor has been launched as part of the ARIES-CS study. On the physics side, it involves adapting and using codes to better assess the location of the divertor and to estimate the corresponding heat loads. Details of the physics study are provided in Ref. 20. On the engineering side, the effort focused on evolving a design well suited to the compact stellarator with the capability to accommodate a heat flux of 10 MW/m^2 (as a reasonable initial goal in anticipation of the physics modeling results) and that can be integrated with the in-reactor component design. In addition, the alpha-particle loss fraction is fairly high, ~5%, and concerns exist as to the accommodation of the resulting maximum heat load and particle flux on the armor. Recent results indicate that most of the alpha loss energy would be deposited in the divertor region.[20] The overall divertor analysis was performed assuming that all the alpha loss energy is also deposited in the divertor region, with a total coverage of ~10.6%, including a transition region from the divertor plates to the first wall.

VI.A. Divertor Configuration and Analysis

The proposed divertor configuration consists of a "T-tube" illustrated in Fig. 18 and described in detail in

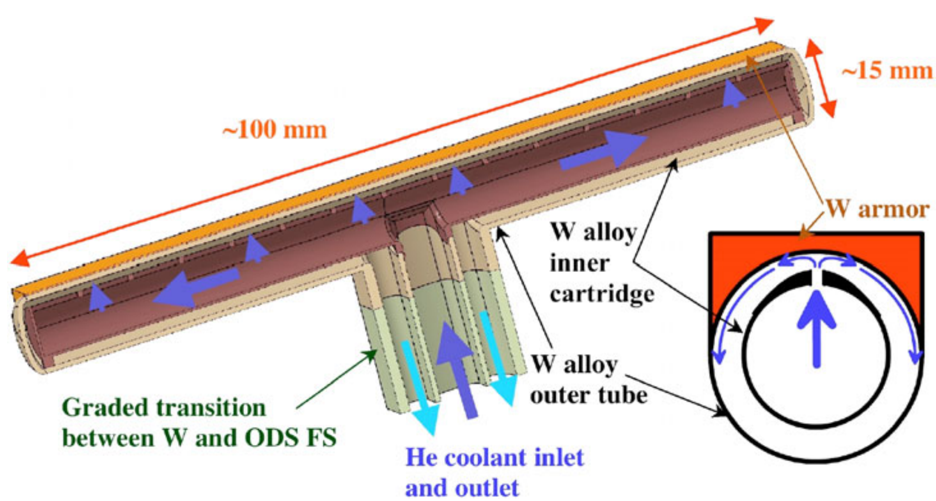

Fig. 18. Cut of the divertor T-tube unit showing the different components and the He flow configuration.

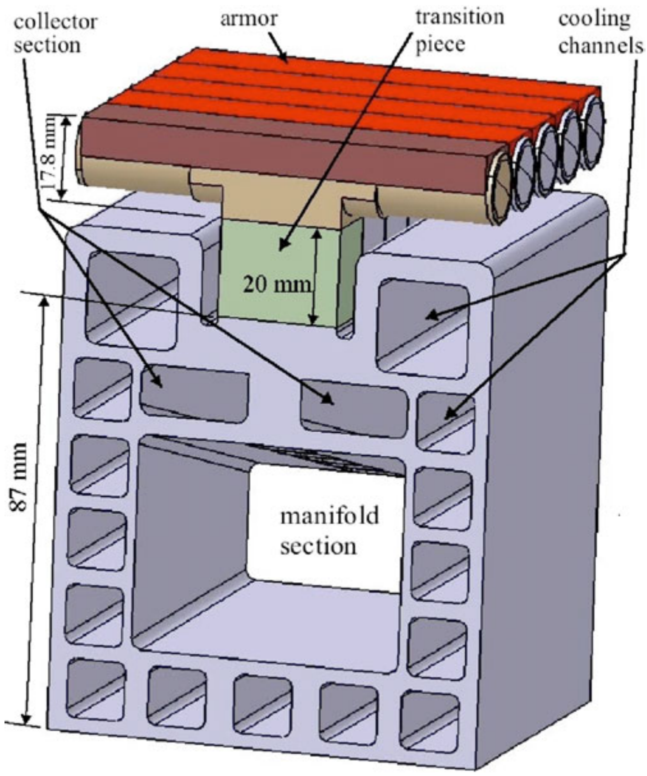

Fig. 19. Divertor manifold unit consisting of several T-tubes on a manifold section (the divertor target plate consists of the assembly of several such units).

Ref. 21. The T-tube is ~15 mm in diameter and ~100 mm long and is made up of a W alloy inner cartridge and outer tube on top of which would be attached a W armor layer subject to the plasma heat flux. The separately fabricated inner cartridge is inserted inside the outer tube and caps are welded at each end. Possible manufacturing techniques for the W alloy components include plasma spray and chemical vapor deposition. Both W alloy pieces are connected to a base ODS FS unit through a graded transition (e.g., using diffusion bonded layers or plasma spray of graded W alloy/ODS FS composition) to minimize thermal stresses. The design provides some flexibility in accommodating the divertor area since a variable number of such T-tubes can be connected to a common manifold to form the desired divertor target, as shown in Fig. 19.

As illustrated in Fig. 18, the helium coolant is routed through the inner cartridge first and is then directed through thin slots (~0.4 mm) to cool the heat-loaded outer tube surface. A 2-D-shaped impinging slot jet is created, leading to high heat transfer at reasonable pressure drop. After impingement, the coolant flows as a highly turbulent wall jet along the large inside surface of the tube and then returns in the lower section of the annular gap between tube and cartridge. Typical parameters for the divertor are summarized in Table V.

TABLE V

Summary of ARIES-CS Divertor Parameters

Divertor T-tube unit dimensions	9 cm (toroidal) × 1.6 cm (poloidal)
Divertor surface coverage	10.6%
Typical divertor plate dimensions	~3.25 m (toroidal) × ~1 m (poloidal)
Number of divertor plates	24
He inlet temperature	573°C
He outlet temperature	700°C
He inlet pressure	10 MPa
Plasma-side annular channel thickness	0.5 mm
He jet velocity	200 m/s
Average jet flow h	~17 000 $W/m^2 \cdot K$
He pressure drop	0.45 MPa
Conducted power to divertor	23.6 MW
Alpha loss power to divertor	23.6 MW
Radiated power to divertor	48 MW
Thermal power removed by He (including 23.6 MW of alpha loss power and 24 MW of friction power)	186 MW
Total mass flow rate	283 kg/s
Pumping power	~27 MW
Maximum W alloy temperature	<1300°C
Maximum primary + secondary stresses	<370 MPa

The thermal power in the divertor is consistent with the assumption that all alpha loss power ends up on the divertor and with the radiation assumptions discussed in Sec. II. To maintain the maximum heat flux ≤10 MW/m^2 for the power flows to the divertor shown in Table V, the peaking factor on the conducted power and alpha loss power to the divertor should be ≤15.6. The resulting heat flux footprint is shown in Fig. 20 assuming a Gaussian distribution. From the detailed analysis performed in Ref. 21, the T-tubes can be configured as a divertor plate with a required He flow rate of ~6 kg/s per m^2 of divertor plate surface area under the heat flux distribution shown in Fig. 20 for zones 2, 3, and 4 over a fractional poloidal distance of ±0.1 (~20% of the divertor area). Zones 1 and 5 have a very low heat load and would require virtually no increase in mass flow rate if the He is flown in series. Due to the uncertainty in the footprint location in particular with the presence of the alpha loss power, it was decided to conservatively assume that the divertor target should be able to accommodate the zones 2, 3, and 4 footprint over 60% of its area. The corresponding mass flow rate was calculated and used to obtain the overall He thermal-hydraulic parameters for the divertor listed in Table V.

The inlet and outlet He temperatures are ~570 and ~700°C, which fit within the overall heat exchanger and Brayton cycle scheme. A stress analysis was also performed, indicating that the total stress intensity (primary and secondary stresses) is <370 MPa for the entire geometry, which is assumed to be less than the $3S_m$ limit

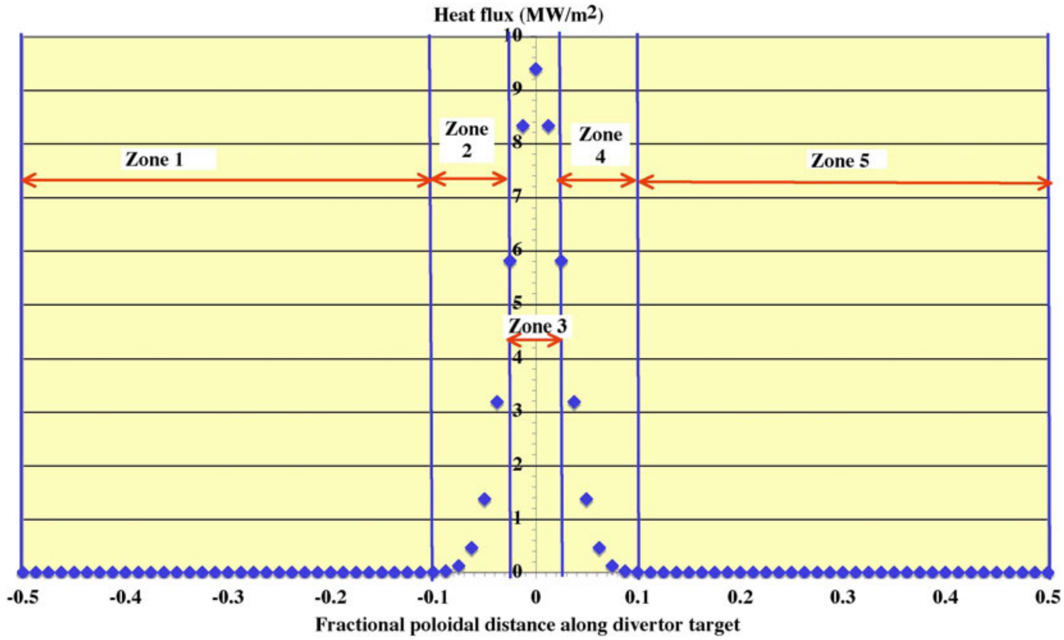

Fig. 20. Gaussian heat flux footprint assumed for divertor plate.

of an anticipated W alloy at the corresponding temperatures.[22] However, development of such a W alloy is a key R&D issue. The thermal-hydraulic performance of the divertor T-tube was confirmed through detailed thermal-hydraulic modeling and experimental studies, described in Ref. 22.

VI.B. Divertor Integration in Power Plant

The divertor plates are exchanged through the main maintenance ports. There are eight divertor plates per field period, and each plate has average dimensions of 3.25 m (toroidal) × 1.0 m (poloidal). Each divertor plate is attached directly to the colder vacuum vessel, with the divertor coolant supply tube/mechanical attachment penetrating the breeding modules and the hot shielding ring, as illustrated in Fig. 21. In this case the concentric coolant pipes to each of the divertor plates are attached at the back of the vacuum vessel and can be cut/rewelded with in-bore tools inserted from the outside of the vacuum vessel. During maintenance,[8] the shield ring, shield block, and inner tube, which are structurally connected, can then be removed as a single unit and in-bore tools inserted for pipe cutting/rewelding. The shield inserts provide the required protection for lifetime components at the back and for rewelding. The cutting and rewelding are done in two steps, with the first cut in the smaller tube near the divertor plate and the second at the back of the larger tube, where the shielding would allow for rewelding (less than ~1 at. ppm He). In addition, if required, this scheme provides the possibility of in situ divertor plate alignment by operation of the positioning screws at the back of the divertor tube, as shown in Fig. 21.

There are a total of 12 openings through the blanket/shield and manifold for divertor pumping, shown in Fig. 5. The divertor pumping ducts are rectangular in shape with dimensions of 0.42 m (radial) × 1.0 m (toroidal) (consistent with past ARIES experience[23]). Neutron streaming is a major concern in the divertor region. Shielding blocks and rings are used inside the divertor access tube but also behind the divertor plates and around penetrations in order to protect the magnet and make the vacuum vessel and blanket manifold lifetime components. The shielding studies are described in Ref. 4.

VI.C. Alpha Loss

The alpha loss is estimated at ~5% for the ARIES-CS case. This is important since it not only represents a loss of heating power in the core but, more important, it adds to the heat load and particle flux on the plasma-facing components. Recent studies suggest that most of these alphas will end up on the divertor plates, which would have to accommodate the resulting heat flux in combination with the divertor heat loads.[20] Of additional concern is the possible erosion due to the He flux, in particular any exfoliation that could result from the accumulation of He atoms in the armor. A possible solution would be to use an engineered W armor with a low-porosity nanostructure in the alpha incident regions, which could enhance the release of He. Clearly, this is a key issue for a compact stellarator, which needs to be further studied to

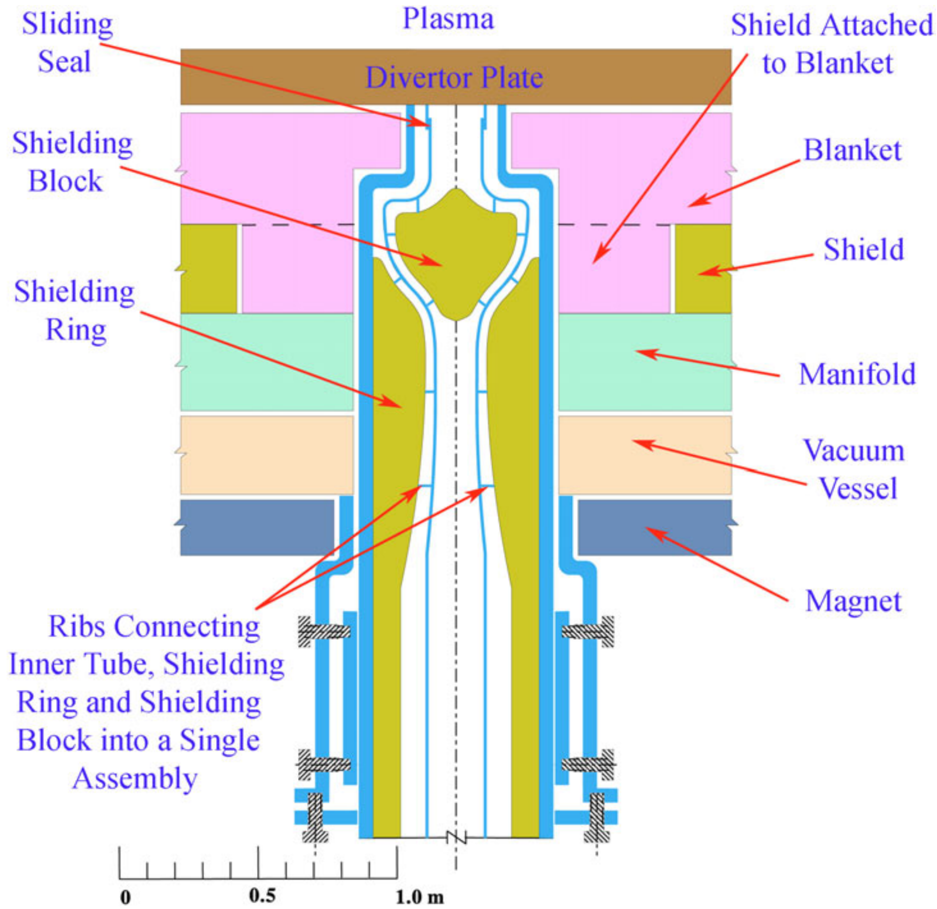

Fig. 21. Schematic of divertor plate coolant supply and attachment.

make sure that a credible solution exists in terms of the alpha physics, the selection of armor material and configuration, and better characterization of the He behavior under prototypic conditions.[24]

VII. COIL DESIGN

The complex 3-D geometry of the coil configuration in a compact stellarator introduces engineering constraints affecting a number of design features such as the distance between plasma and coil, the maximum coil bend radius, the coil support requirement, and assembly and maintenance. The preferred superconductor is Nb_3Sn, operating at $\sim$4 K. It is installed with a wind and react method and heat treated. JK2LB, with yield and ultimate strengths at 4 K of 1420 and 1690 MPa, respectively, is used as structural material.[25]

The overall coil system consisting of the intercoil structure, coil cases, and winding packs is enclosed in a common cryostat. The coils are wound into grooves at the inside of a strong supporting toroidal tube for each field period, as shown in Fig. 22. The supporting tubes are then connected to each other using a bolt arrangement to provide a continuous ring structure to react the large overall centering forces pulling each coil toward the center of the torus, as illustrated in Fig. 5. The out-of-plane forces acting between neighboring coils inside a field period are reacted by the intercoil structure, while the weight of the cold coil system is transferred to the foundation by about three legs per field period. These legs are long enough to keep the heat ingress into the cold system within a tolerable limit.

Magnetic forces and stress analyses were performed using ANSYS to characterize the stress and displacement of the coil structure and to help reduce unnecessary structure in regions where the stresses are low in order to minimize costs. These are described in detail in Ref. 25. The coil structure configuration that would accommodate the structural material stress limits ($\sim$900 MPa) requires average thicknesses of about 20 cm for the intercoil structure and of about 28 cm for the coil strong-back behind the winding pack, as illustrated in Fig. 23.

The possibility of utilizing an advanced, automated fabrication process [based on "additive manufacturing"

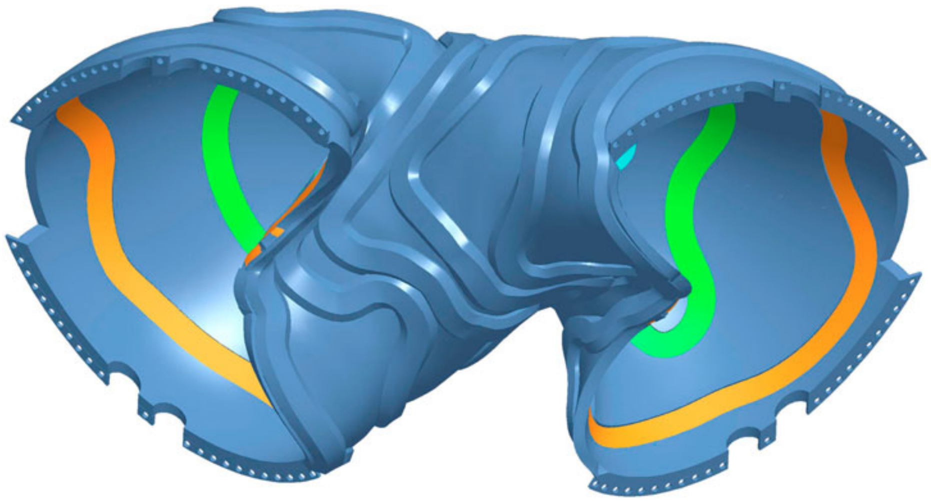

Fig. 22. Coil structure and winding for one field period.

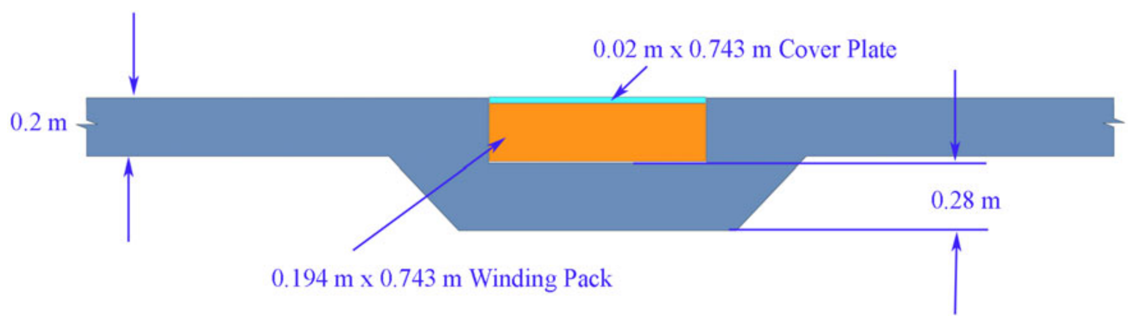

Fig. 23. Average dimensions of coil structure.

to create unique shapes directly from the computer-aided design (CAD) definition file] to reduce the overall coil cost was assessed and is discussed in Ref. 26.

VIII. SAFETY ANALYSIS

Safety studies were performed to demonstrate that the high-level requirements of the U.S. Department of Energy Fusion Safety Standard are met by ARIES-CS, namely, that a site evacuation plan is not needed and that the radioactive waste shall be minimized. The results indicate that through the use of reduced activation materials, a passive decay heat removal system, and the defense-in-depth confinement strategy, the radioactive source terms of ARIES-CS are contained during the design basis accidents analyzed to the degree that no site evacuation plan is needed for ARIES-CS. These radioactive source terms include tritium on cryopumps and implanted into the plasma-facing components (PFCs), activated dust generated by PFC erosion, and ^{210}Po and ^{203}Hg produced by irradiation of the Pb-17Li breeder. For all accidents analyzed, except the divertor loss of coolant/flow accident, the temperatures reached by the in-vessel components of ARIES-CS did not exceed levels that would prevent their reuse upon recovery from these accidents.

The defense-in-depth confinement strategy is based on establishing multiple radioactive confinement barriers between the radioactive source terms in the ARIES-CS vacuum vessel and the environment. For ARIES-CS these barriers are the vacuum vessel, cryostat, heat transport system vault, and auxiliary rooms that adjoin the cryostat. This confinement strategy was also found to provide enough safety margin during a severe bypass accident scenario that the dose to the maximum exposed individual at the site boundary is less than the 10-mSv dose limit for public evacuation, even during conservative weather conditions.

Analyses were also done to minimize the production of waste that fails to qualify for class C, or low-level land burial, through the selection of reduced or low-activation structural and breeding materials for ARIES-CS. The results indicate that the internal components of ARIES-CS will have to be recycled or disposed of in low-level waste repositories but that portions of the magnets, cryostat, and bioshield could meet the national and international clearance criteria for release to private industrial markets.

A detailed description of the safety studies can be found in Ref. 15.

IX. CONCLUSIONS

The ARIES-CS study has developed compact stellarator configurations for power plant application with low plasma aspect ratios and of sizes comparable to advanced tokamak designs. The effort included the development of credible modular coil support, maintenance, and assembly schemes that would accommodate the geometric complexity and the constraints of the maximum allowable field, desirable coil-plasma and coil-coil spacings, and other coil parameters. The final phase of the study focused on a three-field-period configuration with port-based maintenance, field-period–based coil structure unit, a dual-coolant Pb-17Li blanket, and a high-performance He-cooled T-tube divertor configuration.

The design point was pushed to the limit for a "compact" configuration with low aspect ratio to better understand the constraints but with the understanding that in the end it might be better to relax some parameters (e.g., major radius) to provide more margins on space and material stress/temperature limits. Assembly and maintenance and penetration shielding were found to be major factors in the configuration optimization because of geometry and space constraints associated with a compact stellarator. Integration is particularly important because of interfaces and the mutual impact of changes in one system design on others, including modular coil design and structural support, power core design, and maintenance and assembly.

Alpha loss can be appreciable (~5%). Studies indicate that most of the alpha particles will end up on the divertor, which has to be designed to accommodate the combination of alpha loss and divertor heat loads. However, He implantation is a concern since it could lead to armor exfoliation and failure. This is a key issue requiring focused R&D to find an engineering solution (perhaps with a porous nanostructured W armor).

Material development is another key R&D issue. It includes the development of ODS ferritic steel to allow for high-temperature operation (~700°C or more) with acceptable allowable stress (S_m) values under fusion-relevant fluences (~15 MW·yr/m^2) and with demonstrated joining methods. Both the divertor and blanket designs rely on the successful development of such a material. It also includes the development of low-activation W alloy with acceptable properties for use as structural material in the divertor and with demonstrated fabrication and joining methods.

Overall, the engineering effort yielded some interesting and new evolutions in power core design, including the following:

1. blanket/shield optimization to minimize plasma-to-coil minimum distance and reduce machine size[4]
2. separation of hot core components from colder vacuum vessel (allowing for differential expansion through slide bearings)
3. design of coil structure over one field period with variable thickness based on local stress/displacement. When combined with rapid prototyping fabrication technique, this can result in significant cost reduction.[25,26]
4. midsize divertor unit (T-tube) applicable to both stellarator and tokamak [designed to accommodate at least 10 MW/m^2 (Ref. 21)]
5. possibility of in situ alignment of divertor if required
6. first ever 3-D modeling of complex stellarator geometry for nuclear assessment using CAD/MCNP coupling approach[4]
7. significant reduction in stellarator radwaste stream.[15]

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